

National University of Computer and Emerging Sciences
Karachi Campus

Software Design and Analysis (CS3004)

Sessional-I Exam

Date: February 24, 2026

Total Time: 1 Hours

Course Instructor(s)

Total Points: 30

Dr. Muhammad Yaseen Khan, Syed Ahmed Khan,
Mahrukh Khan, Muhammad Ahmed Qaiser, Maqsood
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Total Questions: 04

Semester: Spring-2026

Student Name

Roll No

Section

Student Signature

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Instructions

- Attempt all questions in order. Out of order answers will not be graded.
- Examiner will not visit the examination hall. Make appropriate assumptions where required and mention in the answer sheet.
- If you are assuming any case or functionality, which is not written/included in the paper, please write it to avoid any hassle during grading.

Scenario:

You have recently been appointed as the Head of Engineering at a mid-sized software company that has secured a contract to develop a Medicine Sales Management System for a pharmaceutical client. The system aims to digitize the complete medicine sales and distribution process, which is currently handled through manual records and disconnected tools. It will be used by multiple stakeholders: sales representatives who record doctor visits, place categorized medicine orders (urgent or scheduled), and each sales order may contain multiple medicines, track targets, and generate reports; doctors who review product catalogs, request samples, provide forecasts, and submit feedback; distributors who process orders, manage stock, handle deliveries, generate invoices, and verify payments; manufacturers who monitor sales performance, manage product lists and pricing, and analyze forecasts; and system administrators who manage users and system settings. The system must provide secure role-based access, real-time order tracking, inventory alerts, reporting dashboards, notifications, integration with accounting systems, and web/mobile access. Requirements are still evolving, and the team has limited domain knowledge, requiring frequent client interaction.

CLO 3: Apply and use UML to visualize and document the design of software systems.

Question 1:

[12 points]

Draw one complete Use Case Diagram that models this system. Your diagram must:

- a. Clearly show all actors and use cases. [6 points]
- b. Correctly apply the <<include>> relationship to model mandatory functionality. [3 points]
- c. Correctly apply the <<extend>> relationship to model optional behavior. [3 points]

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CLO 2: Design and Implement OOD models and refine them to reflect implementation details.

Question 2

[6 points]

Write the fully dressed use case description for the use case "generate invoice" using the full use case description template. Note: You may make your own assumptions about the exact use case steps.

CLO 2: Design and Implement OOD models and refine them to reflect implementation details.

Question 3

[6 points]

Draw a UML Class Diagram for the scenario:

- a. Identify the classes, their key attributes and operations. [3 points]
- b. Identify the relationships between classes and specify the multiplicities. [3 points]

CLO 1: Understand the role of design and its major activities within the software development process, with focus on the Unified Process.

Question 4

[6 points]

To develop a software according to the above given requirements:

- a. Identify the most appropriate SDLC process model. [3 points]
- b. Explaining at least two reasons why it fits this scenario. [3 points]